Wooseung Kim

COMPUTER VISION RESEARCHER | SEGMENTATION · REGISTRATION · LABEL-EFFICIENT LEARNING

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SUMMARY

Computer vision researcher (M.S., KAIST; expected Feb 2027) building deep learning models for 2D and 3D image analysis under real-world constraints: **scarce or noisy labels**, **tiny and highly imbalanced targets**, and **distribution shift across acquisition devices**. **1st place in both tracks of the MICCAI 2025 TopBrain challenge**; 5 first-author publications and 1 patent application. Works end to end, from data and training pipelines to Dockerized inference.

TECHNICAL SKILLS

Programming & Tools: Python, PyTorch, Docker, Git, Linux
Vision Methods: Semantic Segmentation, Deformable Registration, Image-to-Image Translation, Generative Models, Self-Supervised and Unsupervised Learning
Problem Expertise: Class Imbalance with Small and Thin Targets · Limited and Noisy Labels · Domain Shift

RESEARCH & PROJECT EXPERIENCE

Calibration Scan-Free MRI Distortion Correction via Unsupervised/Self-Supervised Learning 2025 – Present
KAIST MRI Lab *arXiv 2026* | *ISMRM 2026*

- **Removed the need for an extra calibration scan:** unsupervised distortion correction using only routinely acquired images.
- Designed an invertible-network image translator that turns cross-modal alignment into a same-modality problem, enabling label-free training of a diffeomorphic registration network, combined with test-time instance optimization.
- **Outperformed published baselines on NRMSE, PSNR, and SSIM**, both in-distribution and on two **unseen-vendor datasets** (trained on GE scanners; tested on Philips and Siemens).
- Extending to a **single-forward-pass self-supervised model** for diffusion MRI using physics-informed field augmentation, trained on one vendor and tested on three without fine-tuning or per-case optimization (manuscript in preparation).

Multi-Class Brain Vessel Segmentation – 1st Place, MRA & CTA Tracks Sep 2025
TopBrain 2025, MICCAI · Team KDH · Team leader · 15 competing teams *Certificate* | *Challenge paper* | *Docker image*

- Segmented **over 40 vessel classes**, many thin and small, in 3D MRA and CTA scans from only **25 annotated cases** per modality.
- For MRA, scaled beyond the labeled set with **masked autoencoder (MAE) self-supervised pretraining**, about 1,000 public scans, and **noisy pseudo-labels** on 100 unlabeled scans; trained a five-fold nnU-Net (ResEnc-UNet) ensemble with skeleton-recall loss to keep thin vessels connected.
- **Cut false positives** with a dedicated binary model filtering the multi-class output; 81.4% Dice and 86.3% clDice on the hidden MRA test set.
- **Shipped inference as a Docker container** within platform limits (one 16 GB T4 GPU, at most 15 minutes per volume, container image under 10 GB); included in the organizers' public Docker release.

Sparse Small-Structure Segmentation – 3rd Place Oct 2024
EPVS 2024 Challenge, MICCAI · Team HufsAIM · Lead method developer *Certificate* | *Challenge paper*

- Targeted enlarged perivascular spaces: **tiny, sparse, irregular structures with extreme foreground-background imbalance**.
- Built a **multi-view ensemble** of four SwinUNETR models (three 2D models for axial, coronal, and sagittal views plus one 3D model) with combined Dice and cross-entropy loss and probability averaging to favor small-target recall; **highest instance-level recall among all teams** in the validation phase (centrum semiovale region).

Thin-Structure Segmentation with Label-Preserving Augmentation 2024 – 2026
HUFS AIMLAB · clinical collaboration with radiologists *NMR in Biomedicine 2026* | *iMRI 2024* | *ISMRM 2025* | *Code*

- Designed a **label-preserving augmentation** that applies each geometric transform and its inverse to the image only, preventing the **label noise** that mask resampling introduces on thin structures.
- **Improved robustness on an unseen external dataset** (Dice from 0.847 to 0.919, IoU from 0.738 to 0.852) with on-par internal accuracy, from only 50 annotated cases; added a subregion-specific topology loss for thin-tube connectivity.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Mar 2025 – Feb 2027 (expected)
M.S. in Bio and Brain Engineering

Hankuk University of Foreign Studies (HUFS) Mar 2018 – Feb 2025
B.S. in Biomedical Engineering

PUBLICATIONS

Journal Articles

- Deep Learning-Based Inner Ear Subregion Segmentation in 3D T2-Weighted MRI Using Label-Preserving Data Augmentation
  Wooseung Kim, Yeonah Kang, Seokhwan Lee, Ho-Joon Lee, and Yoonho Nam
  NMR in Biomedicine, vol. 39, no. 4, e70266, 2026.
- Label-Preserving Data Augmentation for Robust Segmentation of Thin Structure in MRI
  Wooseung Kim, Yeonah Kang, Seokhwan Lee, Ho-Joon Lee, and Yoonho Nam
  Investigative Magnetic Resonance Imaging (iMRI), vol. 28, no. 3, pp. 107–113, 2024.

Conference Proceedings

- SACRED: Susceptibility Artifact Correction without Reverse phase-encoding for EPI using Deep learning
  Wooseung Kim and Sung-Hong Park
  Proceedings of the International Society for Magnetic Resonance in Medicine (ISMRM), 2026.

- Deep Learning-Based Topology-Preserving Inner Ear Subregion Segmentation in MRI
  Wooseung Kim, Dayeon Bak, Yeonah Kang, Ho-Joon Lee, and Yoonho Nam
  Proceedings of the International Society for Magnetic Resonance in Medicine (ISMRM), 2025.

Preprints and Submitted Manuscripts

- Unsupervised Susceptibility Distortion Correction of EPI without Calibration Scans via Image Translation-Based Registration
  Wooseung Kim and Sung-Hong Park
  arXiv preprint arXiv:2606.21588, 2026. Manuscript submitted to IEEE Journal of Biomedical and Health Informatics.
- Virtual 1-hour delayed post-contrast 3D heavily T2-weighted FLAIR from 10-minute delayed post-contrast scans using a deep learning model: preliminary study for inner ear application
  Yeonah Kang, Woojin Chung, Seokhwan Lee, **Wooseung Kim**, Yoonho Nam, and Ho-Joon Lee
  Submitted to NMR in Biomedicine, 2026.

Patents

Apparatus and Method for Automatically Measuring Diagnostic Assistive Indicators of Thyroid Eye Disease Using Medical Images
Korea Patent Application No. 10-2024-0172871 Nov 2024

Awards and Honors

1st Place – TopBrain 2025 Challenge (Multi-Class Brain Vessel Segmentation) Sep 2025
International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) 2025 *Certificate*

- 1st place on both the magnetic resonance angiography (MRA) and computed tomography angiography (CTA) tracks

International Society for Magnetic Resonance in Medicine (ISMRM) 2025 Scholarship 2025
Sponsored by the Korean Society of Magnetic Resonance in Medicine (KSMRM)

3rd Place – EPVS 2024 Challenge (Enlarged Perivascular Space Segmentation) Oct 2024
International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) 2024 *Certificate*

Teaching Experience

Teaching Assistant – KAIST, BiS452: Biomedical Imaging Sep 2026 – Present
Teaching Assistant – KAIST, BiS499: Machine Learning for Medical Image Analysis Mar 2026 – Jun 2026
Teaching Assistant – KAIST, BiS452: Biomedical Imaging Sep 2025 – Dec 2025

Military Service

Completed mandatory military service, Republic of Korea Jun 2020 – Dec 2021